

Table 2Summary of correlations used to construct a complete saturated pool boiling curve ($\Delta T_{\text{sub}} = 0$) for cryogenics.

Reference (s)	Region/Point	Correlation(s)	Application Range	Remarks
[17–21]	Natural convection (NC) region	$h_{nc} = \frac{k_f}{L_x} C R a_L^n$ $R a_L = \frac{L_x^3 \rho_f (\rho_f - \rho_{f,w}) g (\mu_f c_{p,f})}{\mu_f^2 k_f}$	$\Delta T_{\text{sat}} \leq \Delta T_{\text{sat,ONB}}$	Values for C and n are provided in Table 1
-	Onset of nucleate boiling (ONB) point	-	$\Delta T_{\text{sat}} = \Delta T_{\text{sat,ONB}}$	Intersection point of NC and NB
[12]	Nucleate boiling (NB) region	$h_{nb} = 13.3 q^{*0.665} (1 + 0.52 p^*)^{4.7} Pr_f^{-1.09} \times \left[\frac{1 + 68e^{20(p^*-1.1)}}{1 + 0.0045 e^{(q^* \times 10^{-5})}} \right]$	$\Delta T_{\text{sat,ONB}} < \Delta T_{\text{sat}} < \Delta T_{\text{sat,CHF}}$	
[13]	Critical heat flux (CHF) point	$\Delta T_{\text{sat,CHF}} = T_{\text{sat,CHF}} - T_{\text{sat}} = \frac{q_{\text{CHF}}}{h_{\text{CHF}}}$	$\Delta T_{\text{sat}} = \Delta T_{\text{sat,CHF}}$	
[15]		$q_{\text{CHF}}^* = [0.16 - 0.104 (p^*)^{10}] \times [1 - 0.004(p^*)^{\theta} \cos(\left(\frac{88}{180}\right)\theta) ^{0.364} \times [1 + 0.16 \left(\frac{c_{p,f} \Delta T_{\text{sub}}}{h_{f,g}}\right)]] \times \left[\rho_g h_{f,g} \left(\frac{\sigma g (\rho_f - \rho_g)}{\rho_g^2}\right)^{\frac{1}{4}} \right]$		
[13]	Transition boiling (TB) region	$\Delta T_{\text{sat,tb}} = [\Delta T_{\text{sat,CHF}} + \varphi (\Delta T_{\text{sat,min}} - \Delta T_{\text{sat,CHF}})]$ $\varphi = \left \left(\frac{q^* - q_{\text{CHF}}^*}{q_{\text{min}} - q_{\text{CHF}}^*} \right)^{0.8} \right $	$\Delta T_{\text{sat,CHF}} \leq \Delta T_{\text{sat}} \leq \Delta T_{\text{sat,min}}$	Accuracy of this correlation is strongly dependent on predictions of the CHF and MHF points
[16]	Minimum heat flux (MHF) point	$T_{w,\text{min}} = \left[T_{\text{sat}} + (T_c - T_{\text{sat}}) \left(-9.1 + 12 \left(\frac{k_f \rho_f c_{p,f}}{k_w \rho_w c_{p,w}} \right)^{0.025} \right) \right]$	$\Delta T_{\text{sat}} = \Delta T_{\text{sat,min}}$	This correlation accounts for surface material effect
[16]		$q_{\text{min}}^* = 0.043 \left[\frac{c_{p,g} k_g^2}{\mu_g \rho_g g (\rho_f - \rho_g)} \right]^{0.567} \times (-0.107 + 0.38 (\Delta T_{\text{sat,min}})^{0.39})^{3.094}$		This correlation is dependent on $\Delta T_{\text{sat,min}}$ correlation
-	Blending function between q_{min}^* and $1.5q_{\text{min}}^*$	$\Delta T_{\text{blend}} = \Delta T_{\text{sat,min}} + \left \frac{q^* - q_{\text{min}}^*}{0.5q_{\text{min}}^*} \right ^{0.5} \times (\Delta T_{\text{sat,tb}} _{1.5q_{\text{min}}^*} - \Delta T_{\text{sat,min}})$	$\Delta T_{\text{sat,min}} \leq \Delta T_{\text{sat}} \leq \Delta T_{\text{sat,tb}} _{1.5q_{\text{min}}^*}$	FB correlation is terminated at $1.5q_{\text{min}}^*$ and the blending function is used to merge the FB curve with the MHF point
[14]	Film boiling (FB) region	$h_{fb} = (0.148 + 0.052 \sin \theta) \times \frac{k_g}{L_b} \left[R a_g \left(\frac{h_{f,g} + 0.46 c_{p,g} (T_{w,fb} - T_{\text{sat}})}{c_{p,g} (T_{w,fb} - T_{\text{sat}})} \right) \right]^{0.33} + \frac{(1.7 - 0.55 \sin \theta) \sigma_b (T_{w,fb}^4 - T_{\text{sat}}^4)}{(T_{w,fb} - T_{\text{sat}})}$	$\Delta T_{\text{sat}} \geq \Delta T_{\text{sat,tb}} _{1.5q_{\text{min}}^*}$	Correlation also accounts for radiation effects

points, followed by the $T_{w,\text{CHF}}$ correlation, 10.71 % for $N = 236$, h_{fb} correlation, 12.94 % for $N = 1209$, h_{tb} correlation, 14.84 % for $N = 133$, q_{min}^* correlation, 22.42 % for $N = 158$, and h_{nb} correlation, 25.36 % for $N = 2908$. Notice that certain correlations are evaluated against more cryogenics than others; this again is based entirely on the availability of data for a given boiling regime or transition points.

3.2. Validation of new methodology for constructing complete pool boiling curve

While each individual correlation has been thoroughly tested against its respective datasets, Fig. 3, it is crucial to validate the complete boiling curve against experimental data to illustrate the robustness and reliability of the new methodology for the construction of complete boiling curve. Such validation is a key step toward demonstrating practical applicability of the methodology across different cryogenics and operating conditions. To this end, the validation is conducted using data for the three most significant and widely used cryogenics in scientific and industrial applications: liquid nitrogen (LN_2), liquid hydrogen (LH_2), and liquid helium (LHe).

Ideally, one should aim to validate the entire boiling curve against experimental data from a single source. This would provide a direct comparison between the predicted boiling curve and experimental results across all boiling regimes for same operating conditions. Unfortunately, such comprehensive datasets are very rare in published literature since pool boiling experiments are typically conducted by different researchers, each focusing on specific regime and operating conditions

based on their own research objectives. As a result, it is difficult to find a single source of experimental data covering the entire boiling curve.

To address this shortcoming, we took data from different studies, each focused on a particular boiling regime for same cryogen, which are then combined to validate the methodology. This allows validation across multiple datasets, which demonstrates broad applicability of the methodology. For example, Fig. 4 shows the validation for LHe using experimental data from five different sources [22–26] at a pressure of 101 kPa. The results demonstrate very good capability in predicting the LHe data, which include both horizontal and vertical surfaces. Validation of correlations for the ONB, CHF and MHF points is deferred to a later section.

However, some discrepancies arise in the natural convection regime, where predictions show some deviations from the data. This divergence is attributed to several factors. First and foremost is experimental uncertainty, particularly at extremely low temperatures. For LHe, wall superheat can be as low as 0.01 K, which would greatly accentuate measurement uncertainties. Another contributing factor could be limitations of the natural convection correlations used. These correlations were originally developed for mostly room-temperature fluids, and their applicability at such low temperatures may be somewhat compromised. Moreover, as shown in Fig. 4 (b), the validation for natural convection over vertical surfaces for LHe is incomplete due to a lack of corresponding experimental data. This data shortfall highlights the need for further experimental studies, particularly for cryogenics like LHe, to provide a more comprehensive database that covers all boiling regimes, including natural convection, over a range of surface orientations.